

1

Question

DO VLMS REALLY LOOK?

- Text pretraining teaches facts such as Adidas having three stripes or chickens having two legs.
- The study modifies familiar images but asks neutral counting or identification questions without adversarial text.
- If the model repeats the canonical fact, prior knowledge has overridden visual evidence.

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Evidence

PRIORS DOMINATE

- Five SOTA VLMs score 100% on originals but only 17.05% mean counting accuracy on counterfactuals.
- 75.70% of responses match the biased answer; Y/N identification collapses to 25.11% on modified images.
- Errors span tiny counts: animal legs 2.12%, board rows/columns 2.26%, patterned grids 22.44%.

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VLMBias

- 1,392 images span animals, logos, flags, chess, game boards, optical illusions and patterned grids.
- Editors and scripts add/remove signature elements; humans reject ambiguous or low-quality examples.
- Original-image sanity checks verify that all five tested VLMs know the subject and canonical count.

VLMBIAS

Vision-Language Models Are Biased

When Prior Knowledge Overrides Visual Evidence | ICLR 2026

Familiar visual contexts trigger memorized facts so strongly that leading VLMs ignore counterfactual details, failing even neutral counting and identification questions.

visual bias

counterfactuals

counting

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Causal clues

BACKGROUND + OVERTHINKING

- Removing familiar backgrounds raises accuracy by 21.09 points and cuts bias rate by 40.58 points.
- More reasoning initially raises accuracy toward about 40%, then overthinking makes it decline.
- Tools help when used longer, but overconfidence means o4-mini invokes them on only 29.66% of questions.

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Metrics

ACCURACY VS BIAS RATE

- Q1/Q2 request objective counts; Q3 asks whether the modified subject is still the original.
- Bias rate measures how often responses match the canonical, memorized answer rather than ground truth.
- A vision-encoder linear probe reaches 95.26% on leg counting, suggesting the visual signal exists before language fusion.

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Implications

- Visual cues can activate language priors that swamp correctly encoded sensory information.
- Larger models may be more biased; explicitly counting-trained models improve to 36.02% but remain imperfect.
- Future systems need calibrated uncertainty, automatic crop/zoom use, anomaly checks and bias-aware visual training.